KAIZEN MATHS

Lesson starter

Binomial and Geometric Distributions

Binomial

Name _____

S1

X ~ B(9, 0.25). Find P(X = 5).

S2

X ~ B(8, 0.25). Find P(X ≤ 3).

S3

X ~ B(15, 0.15). Find P(X ≥ 4).

S4

X ~ B(17, 0.6). Find E(X) and Var(X).

Show your working

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STA-ABDB-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Binomial and Geometric Distributions

Binomial

Name _____

E1

X ~ B(10, 0.55). Find P(X = 7).

E2

X ~ B(13, 0.55). Find P(X ≤ 4).

E3

X ~ B(9, 0.35). Find P(X ≥ 4).

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Binomial

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Show your working

Teacher answers & working

Binomial and Geometric Distributions · Binomial

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S1

X ~ B(9, 0.25). Find P(X = 5).

Answer:

0.0389

Working:

This is a binomial exact probability.

Use $P(X = r) = nCr\ p^r(1 - p)^{n - r}$.

$P(X = 5) = 9C5(0.25)^5(0.75)^4$.

$9C5 = 126$.

$P(X = 5) \approx 0.0389$.

0.0389

S2

X ~ B(8, 0.25). Find P(X ≤ 3).

Answer:

0.8862

Working:

Add the binomial probabilities from 0 up to 3.

$P(X \leq 3) = P(X = 0) + P(X = 1) + ... + P(X = 3)$.

Use $P(X = k) = nCk\ p^k(1 - p)^{n - k}$ for each term.

$P(X \leq 3) \approx 0.8862$.

0.8862

S3

X ~ B(15, 0.15). Find P(X ≥ 4).

Answer:

0.1773

Working:

Use the complement because this is an at-least probability.

$P(X \geq 4) = 1 - P(X \leq 3)$.

$P(X \leq 3) \approx 0.8227$.

$P(X \geq 4) \approx 1 - 0.8227 = 0.1773$.

0.1773

Teacher answers & working

Binomial and Geometric Distributions · Binomial

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S4

X ~ B(17, 0.6). Find E(X) and Var(X).

Answer:

$E(X) = 10.2$, $Var(X) = 4.08$

Working:

For $X \sim B(n, p)$, $E(X) = np$ and $Var(X) = np(1 - p)$.

$E(X) = 17 \times 0.6 = 10.2$.

$Var(X) = 17 \times 0.6 \times 0.4 = 4.08$.

$E(X) = 10.2$, $Var(X) = 4.08$

E1

X ~ B(10, 0.55). Find P(X = 7).

Answer:

0.1665

Working:

This is a binomial exact probability.

Use $P(X = r) = nCr\ p^r(1 - p)^{n - r}$.

$P(X = 7) = 10C7(0.55)^7(0.45)^3$.

$10C7 = 120$.

$P(X = 7) \approx 0.1665$.

0.1665

E2

X ~ B(13, 0.55). Find P(X ≤ 4).

Answer:

0.0698

Working:

Add the binomial probabilities from 0 up to 4.

$P(X \leq 4) = P(X = 0) + P(X = 1) + \ldots + P(X = 4)$.

Use $P(X = k) = nCk\ p^k(1 - p)^{n - k}$ for each term.

$P(X \leq 4) \approx 0.0698$.

0.0698

Teacher answers & working

Binomial and Geometric Distributions · Binomial

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E3

X ~ B(9, 0.35). Find P(X ≥ 4).

Answer:

0.3911

Working:

Use the complement because this is an at-least probability.

$P(X \geq 4) = 1 - P(X \leq 3)$.

$P(X \leq 3) \approx 0.6089$.

$P(X \geq 4) \approx 1 - 0.6089 = 0.3911$.

0.3911